

Introduction

Bridges are essential components of modern transportation in the United States, enabling the movement of people and goods. However, these structures are constantly exposed to harsh environmental conditions such as moisture, temperature fluctuations, and deicing salts. Among these factors, **water is one of the most significant contributors to structural degradation**, as it promotes corrosion and rust in metal components.

The corrosion of these structural elements occurs when steel is exposed to water, initiating reactions that form rust and weakening the integrity of bridges over time. Because of this, **every year over \$13.6 billion is spent repairing water-based corrosion damage to stainless steel bridges** in the United States. Traditional protective coatings act as barriers, but degrade quickly and are therefore not effective in the long-run.



Figure 1: Corrosion on Steel Bridge Surface

Recent research has focused on solving this problem through **hydrophobic coatings, which repel water** and reduce the contact area between steel and the liquid. Specifically, certain chemicals such as fatty-acids and polymers show hydrophobic properties. By minimizing water exposure, these surfaces show potential to reduce corrosion and extend the life expectancy of steel bridge structures.

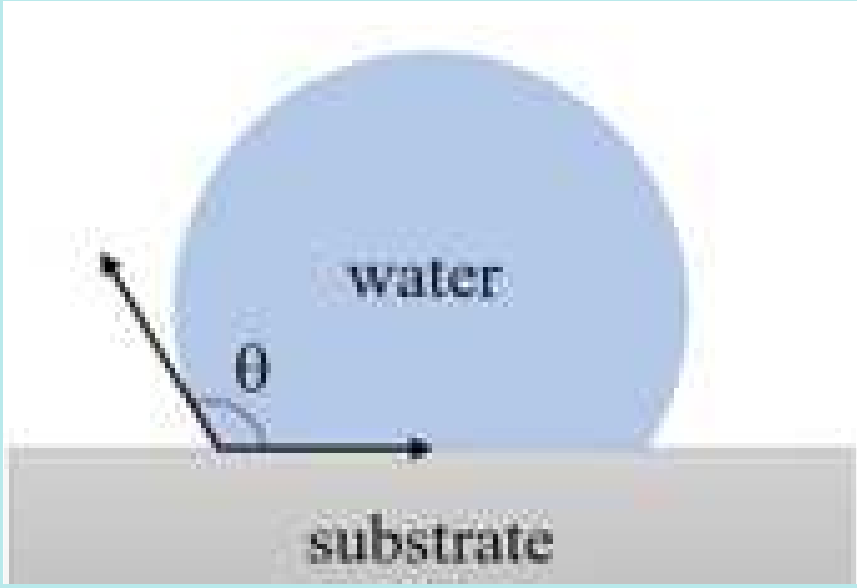


Figure 2: Water Droplet on a Hydrophobic Surface

Purpose

Several hydrophobic chemical coatings made up of polymers, fatty acids, and nanomaterials will be evaluated **to determine if any hydrophobic coatings demonstrate effective water resistance and durability for stainless steel surfaces**. These findings will serve as data to inform cost-effective strategies for extending bridge lifespans and reducing maintenance due to corrosion for bridges in the United States.

HYDROPHOBIC COATINGS FOR PREVENTING WATER DEGRADATION ON BRIDGES

Methods

- 01  Clean stainless steel samples prior to coating them using solvents including acetone, hexene, and silicate abrasive paper.
- 02  Create solutions of 40% wt, 55% wt, and 70% wt stearic acid by combining stearic acid with ethanol. Coat samples in the solutions.
- 03  Create solutions of 4% wt, 12% wt, and 20% wt silicon dioxide nanoparticles (SiNPs) by combining SiNPs with polydimethylsiloxane (PDMS). Coat samples in the solutions.
- 04  Create solutions of 7% wt, 15% wt, and 25% wt tetraethoxysilane (TEOS) by combining TEOS with ammonia and PDMS. Coat samples in the solutions.
- 05  Use a pipette to place a water droplet on each coated surface. Use ImageJ to find the water contact angle on each surface. Then, find the masses of the samples before and after being immersed in salt water for one month. Calculate the % mass lost to corrosion with these values.

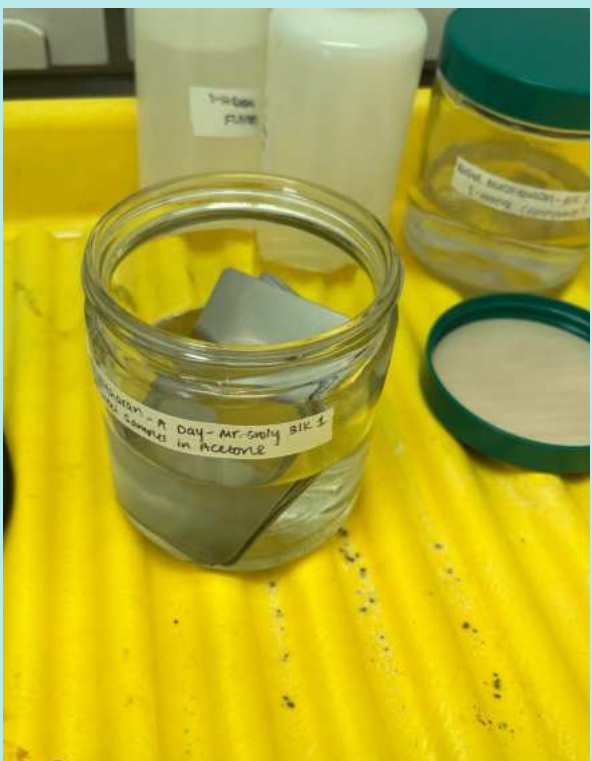


Figure 3: Samples Immersed in Solvents to Remove Contaminants from the Surface

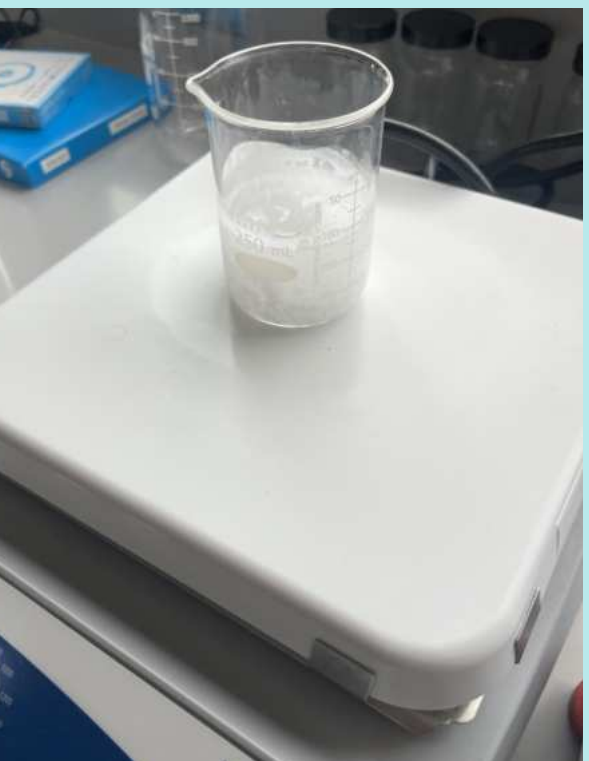


Figure 4: Stearic Acid Dissolving in Ethanol to Create a Hydrophobic Solution

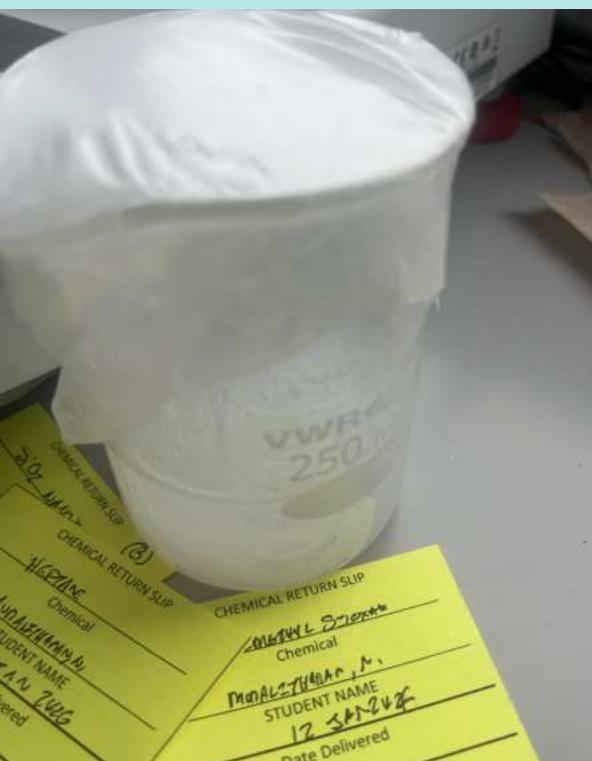


Figure 5: Silicon Dioxide Nanoparticles Immersed in a Polydimethylsiloxane Solution



Figure 6: Tetraethoxysilane Dissolved in Ammonia/ Polydimethylsiloxane Solution



Figure 7: Stearic Acid Coating on Stainless Steel Sample

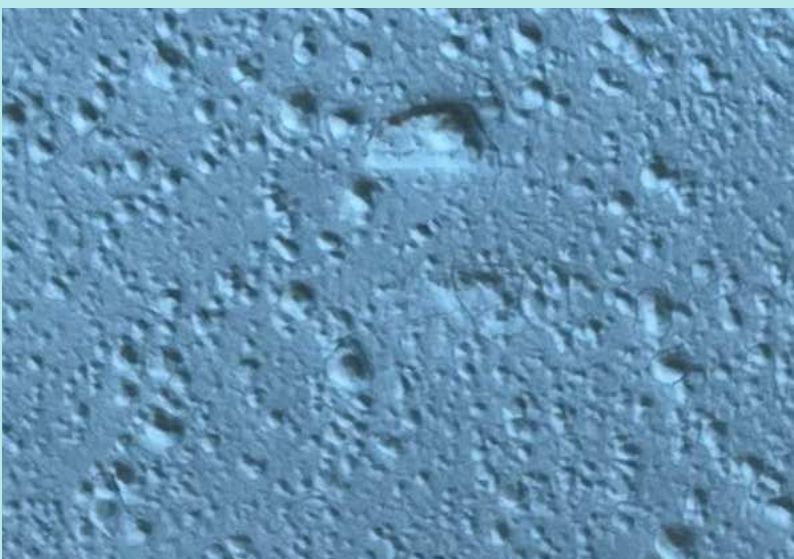


Figure 8: Silicon Dioxide Nanoparticles Coating on Stainless Steel Sample

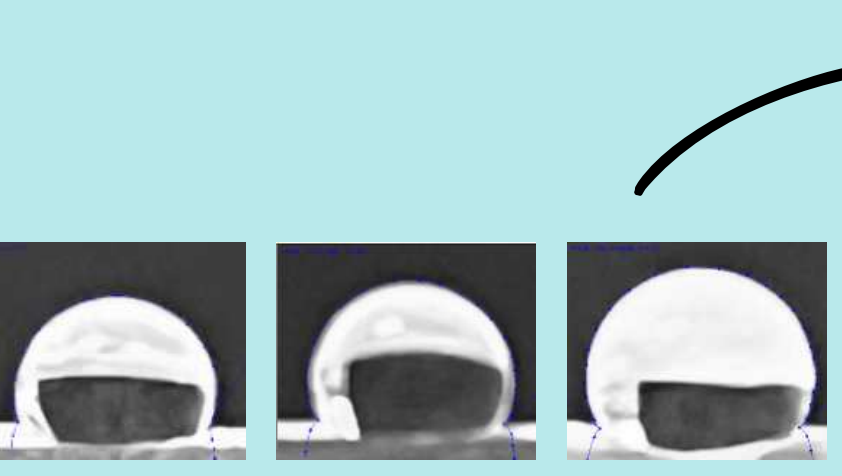


Figure 9: Tetraethoxysilane Coating on Stainless Steel Sample

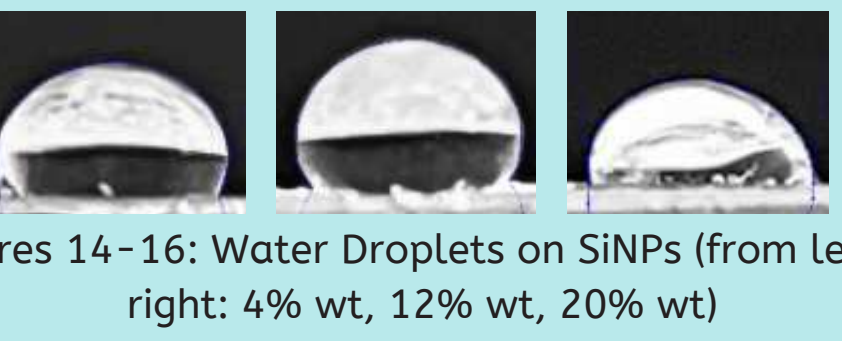


Figure 10: Coated Samples Immersed in Salt Water, Exposed to Sun and Moisture

Results



Figures 11-13: Water Droplets on Stearic Acid (from left to right: 30% wt, 33% wt, 37% wt)



Figures 14-16: Water Droplets on SiNPs (from left to right: 4% wt, 12% wt, 20% wt)



Figures 17-19: Water Droplets on TEOS (from left to right: 7% wt, 15% wt, 25% wt)

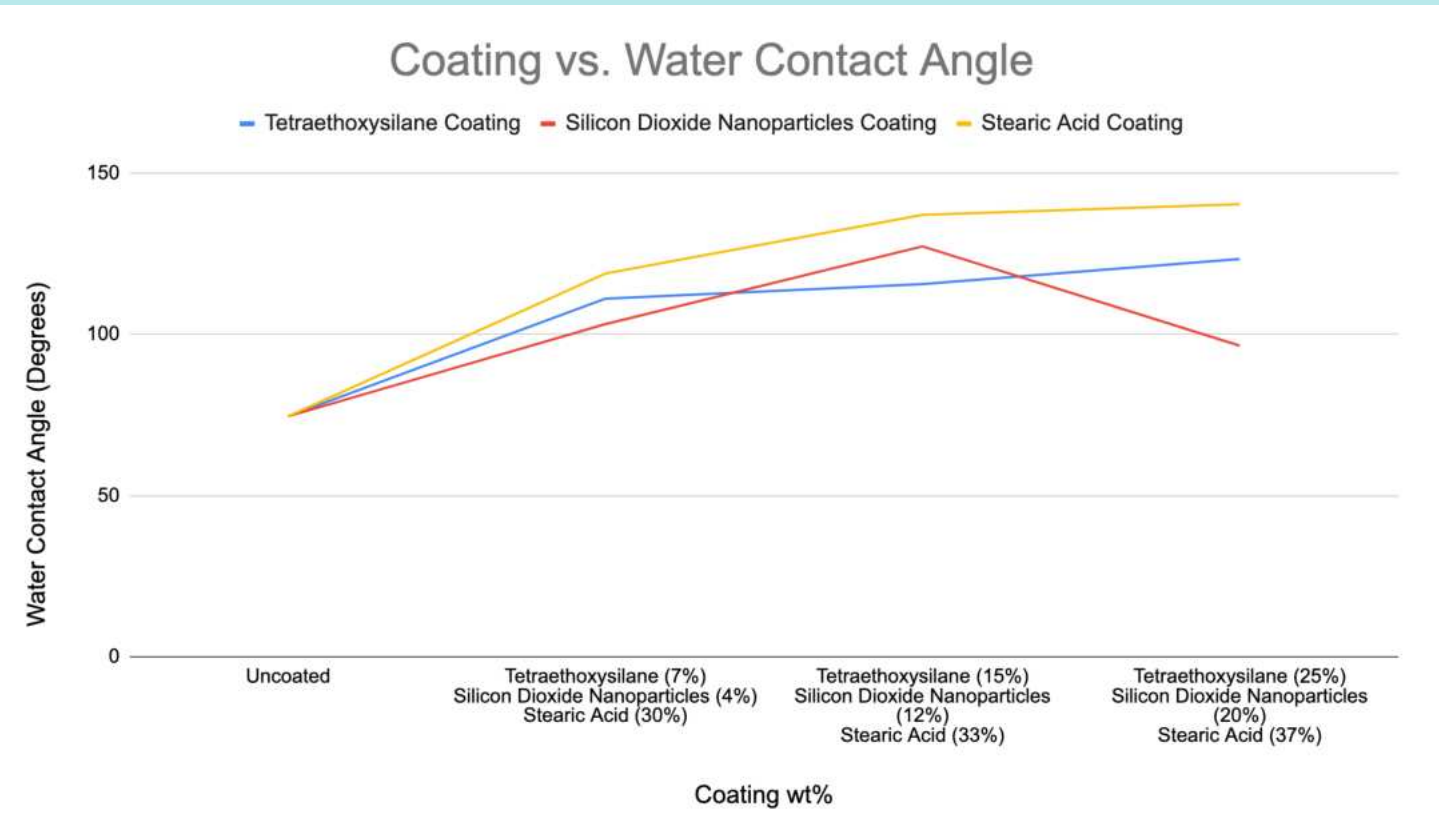


Figure 20: Coating vs. Water Contact Angle Graph

Water Contact Angle (WCA) Results: Stearic acid had the highest WCA (peaking at 37% wt), followed by TEOS (peaking at 25% wt), and SiNPs (peaking at 12% wt).

% Mass Lost to Corrosion Results: Stearic acid had the lowest % mass lost, followed by TEOS, and lastly SiNPs.

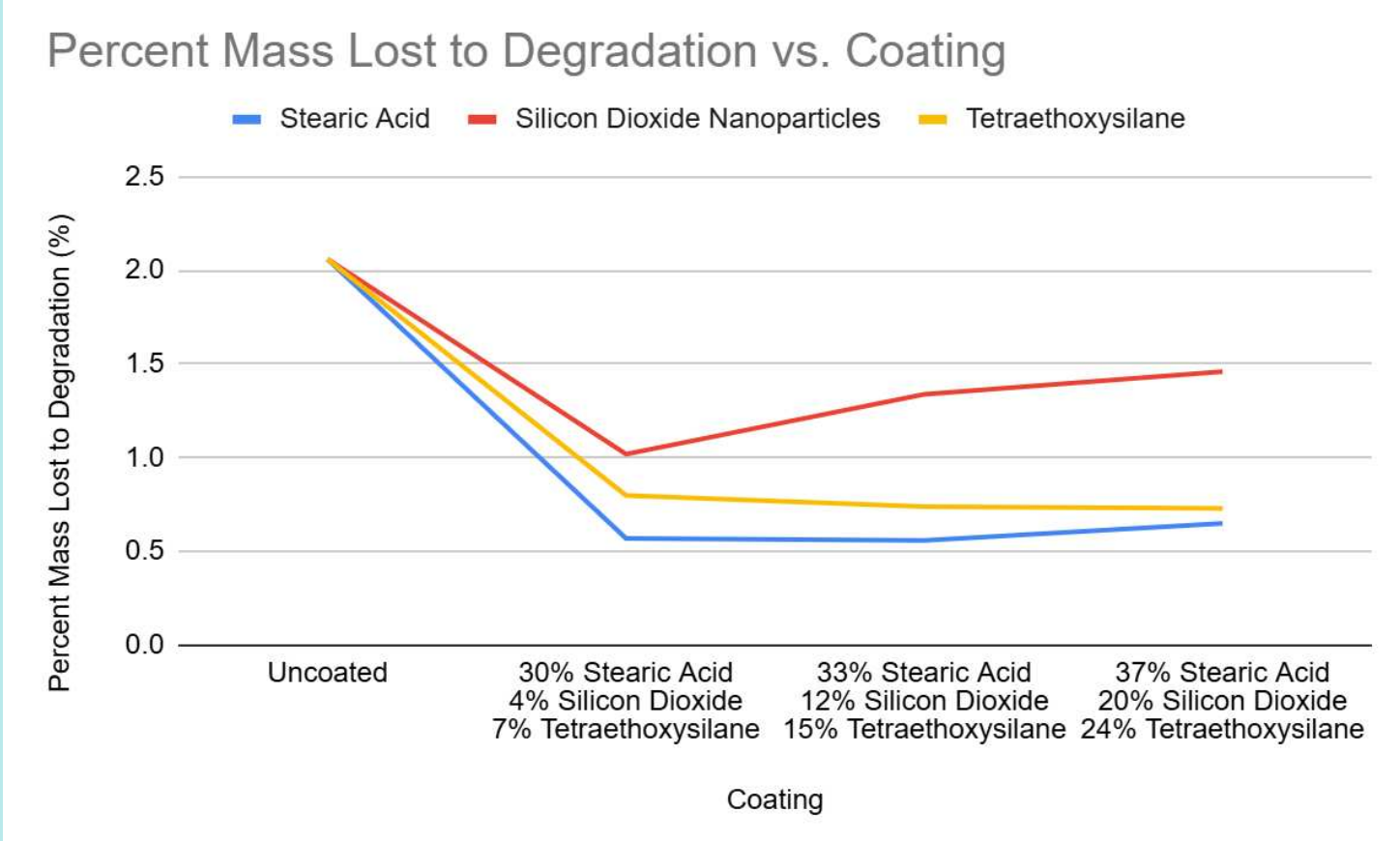


Figure 21: Mass Lost to Degradation vs. Coating Graph

Conclusion

Hydrophobic coatings can significantly reduce water-driven corrosion on stainless steel surfaces by limiting direct contact between water and the metal surface. Among the coatings tested, **stearic acid performed the best, producing the highest water contact angle and the lowest percent mass lost to degradation** after one month of saltwater exposure, which indicates the most effective water repellency and corrosion resistance.



Figure 22: Stearic Acid Coating after Saltwater Exposure Test, Little Rust Observed

TEOS coatings also showed promising results, but they were not as hydrophobic as the stearic acid coatings. Silicon dioxide nanoparticle coatings increased surface roughness, which moderately enhanced the hydrophobicity of the surface, though not as effectively as stearic acid or TEOS. Overall, the results show a **clear relationship between higher surface hydrophobicity and less corrosion, supporting the idea that reducing water contact can slow degradation**. This suggests that simple chemical coatings, particularly **stearic acid, can serve as a cost-effective strategy to extend bridge lifespans and improve infrastructure durability**.

Extensions

Overall, fatty acids such as stearic acid performed the best in terms of water repellency and mitigating degradation. **Future research could explore additional fatty acids** such as caprylic acid, **test long-term durability under more accurate real-world conditions** such as by simulating the influence of moisture fluctuations and deicing salts, and **examine how these coatings perform on different types of metals** including carbon steel to better understand their broader applications.

Citations

Alcántara, J., Fuente, D. de la, Chico, B., Simancas, J., Díaz, I., & Morcillo, M. (2017, April 13). Marine Atmospheric Corrosion of Carbon Steel: A Review. MDPI. <https://www.mdpi.com/1996-1944/10/4/406>

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